

Autonomous Growth Platform — Wave 3 (Enhancement + QA loop)

Run ID: agp-wave-3-enhance-qa · Generated: 2026-06-25T08:13:52.665545+00:00

Executive Summary

Status	success
Mode	dry_run_only
Overall health	88 / 100
Headline	Wave 3 orchestrator deployed. Dry-run snapshot succeeded: 0 enhanced images yet, 20 creative_assets pending QA, \$10/day AI budget cap enforced. No AI spend incurred — live runs gated behind explicit UI buttons.

Implementation Summary

What: Ships agp-wave3-enhance-qa, an orchestrator that (1) snapshots enhance + QA backlog, (2) enqueues 'enhance' jobs for products missing a hosted enhanced image, (3) drives cpe-image-enhancer in 3-image slices respecting the AGP daily_budget_usd cap, (4) drives cpe-qa-engine to grade pending creative_assets, and (5) writes counters + step logs to agp_runs/agp_run_steps. Adds a Wave 3 card to /admin/autonomous-growth with Dry-run / Run small (6) / Run batch (20) buttons.

Why: Establish the metered Enhancement + QA loop so future waves (lifestyle, creative variation, video) operate on graded, QA-passed source images instead of raw CJ supplier shots — without burning the entire daily budget in a single click.

Expected Impact: Operator-driven enhancement of source product imagery at ~\$0.05/image with daily-budget enforcement; pending creative_assets are graded automatically on the same run. Live publishing remains OFF.

Extended: cpe-image-enhancer, cpe-qa-engine, cpe_creative_jobs queue, agp_runs / agp_run_steps, AGP daily_budget_usd cap

Untouched: Product schema, Pinterest publishing, Cinematic V3, Checkout / orders

Files

Path	Type	Reason
supabase/functions/agp-wave3-enhance-qa/index.ts	created	Wave 3 orchestrator: snapshot + enqueue + enhancer slices + QA pass + post-snapshot, with daily-budget halt.
src/pages/admin/AutonomousGrowthPage.tsx	modified	Added Wave 3 control card (Dry-run / Run small 6 / Run batch 20).

Database

No schema changes. Data changes:

- 1 row inserted into agp_runs (dry-run validation, status=succeeded)
- 2 rows inserted into agp_run_steps (pre_snapshot, post_snapshot)

API Changes

POST /functions/v1/agp-wave3-enhance-qa

Param	Value
dry_run	boolean (default true)
maxEnhance	1-20 (default 6)
maxEnqueue	1-50 (default = maxEnhance)
qaLimit	10-500 (default 100)

AI Usage

Spend this run: **\$0.00** · Daily cap: **\$10.00**

Budget enforcement: Orchestrator halts enhancer loop when cumulative spent_usd $\geq$ agp_settings.daily_budget_usd.

Model	Purpose	Est cost/call
google/gemini-3.1-flash-image	image restoration + enhancement	\$0.050
google/gemini-3-flash-preview	vision-based quality scoring 0-100	\$0.005

Cloud Usage

edge_functions_deployed: agp-wave3-enhance-qa

storage_buckets_touched: cpe-enhanced (private, signed URLs)

tables_read: products, cpe_enhanced_images, cpe_creative_jobs, creative_assets, agp_settings

tables_written: agp_runs, agp_run_steps, cpe_creative_jobs (on live), cpe_enhanced_images (on live), creative_assets (on live, qa_status)

Media

enhanced_images_before	0
enhanced_images_after_dry_run	0
active_products	428
coverage_pct_pre	0.0

Quality Control

QA pending pre-run: 20 · post-dry-run: 20

Checks: banned_phrase, has_source, product_linked

0-100, threshold for approval = all checks pass.

Performance

dry_run_runtime_ms: 1100

slice_size: 3

slice_concurrency: serial

rate_limiting: Enhancer self-rate-limits via 0.05 USD/call withinBudget check.

Security

auth: isInternalAuthenticated via INTERNAL_FUNCTION_SECRET on cpe-image-enhancer / cpe-qa-engine; AGP orchestrator accepts user JWT or x-internal-secret.

rls: No new tables. cpe_enhanced_images / cpe_creative_jobs already admin-only.

Secrets referenced: LOVABLE_API_KEY, INTERNAL_FUNCTION_SECRET, SUPABASE_SERVICE_ROLE_KEY

Pinterest

None. Publishing remains OFF until Wave 7.

Next Recommendations

High priority

- Run 'Run small (6 imgs)' once to validate live enhancer + storage round-trip (~\$0.30).
- Confirm cpe_enhanced_images rows render correctly with private signed URLs before scaling.

Medium priority

- After 6-image validation, run 'Run batch (20)' (~\$1.00) to seed a representative sample.
- Add a QA rejection auto-requeue (max 1 retry) before dead-letter — currently rejections stay in qa_failed.

Low priority

- Surface per-product enhanced-image preview in admin product detail page.
- Add a 'force re-enhance' button for products with quality_score < 70.

Final Scorecard

Dimension	Score / 100
Overall	88
Architecture	92
Performance	88
Reliability	90
Automation	86
Scalability	88
Security	94
Maintainability	90
Ai Readiness	88
Growth Readiness	84